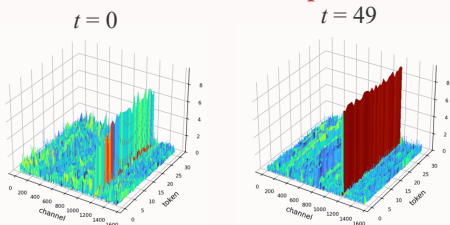


1. The Challenge of DiT Quantization

Activations drift across steps and guidance

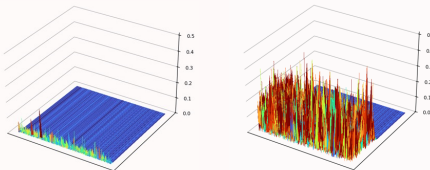
Across timesteps



Across CFG

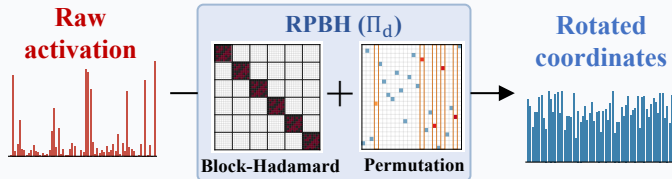
Unconditional

Conditional



2. OrbitQuant Principle

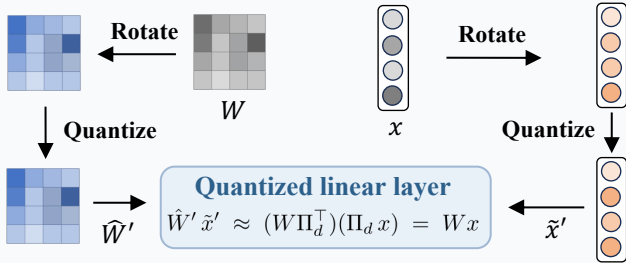
1. Randomized Permuted Block-Hadamard (RPBH)



2. Weight & Activation Quantization

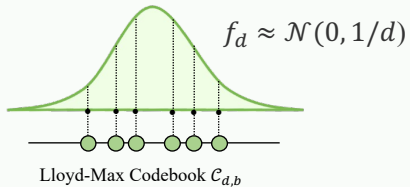
Weights (offline)

Activations (online)

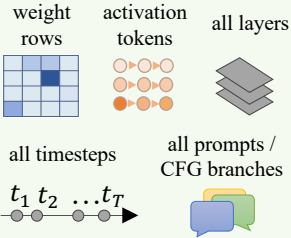


3. Fixed Coordinate Law

Rotation gives one fixed distribution



Same codebook used for



Works across



Image DiTs



Video DiTs

⋮